

Percentage

$$1\% \text{ of } 2000 = \frac{1}{100} \times 2000 = 20 \Rightarrow (20.00)$$

$$10\% \text{ of } 2000 = \frac{10}{100} \times 2000 = 200 \Rightarrow (200.0)$$

• 2345

$$1\% \text{ of } 2345 = 23.45$$

$$10\% \text{ of } 2345 = 234.5$$

	1%	2%	3%	10%	20%	30%
120	1.20	2.40	3.60	12.0	24	36
144	1.44	2.88	4.32	14.4	28.8	43.2
340	3.40	6.80	10.20	34.0	68	102
512	5.12	10.24	15.36	51.2	102.4	153.6
840	8.40	16.80	25.2	84.0	168	252
1152	11.52	23.04	34.56	115.2	230.4	345.6
2016	20.16	40.32	60.48	201.6	403.2	604.8
3240	32.40	64.80	97.2	324.0	648	972
4536	45.36	90.72	136.08	453.6	907.2	1360.8
6025	60.25	120.5	180.75	602.5	1205	1807.5
5045	50.45	100.90	151.35	504.5	1009	1513.5
9900	99.00	198	297	990.0	1980	2970

Approach 1Approach 2

$$64\% \text{ of } 125 = (4\% \times 16) \text{ of } 125$$

$$= \frac{16}{25} \times 125 = 80$$

$$\left(\frac{64\%}{4}\right) \text{ of } (125) \times 4$$

$$= \frac{16}{100} \times 500 = 80$$

$$72\% \text{ of } 250 = (4\% \times 18) \text{ of } 250$$

$$= \frac{18}{25} \times 250 = 180$$

$$\left(\frac{72\%}{4}\right) \text{ of } (250) \times 4$$

$$= \frac{18}{100} \times 1000 = 180$$

$$45\% \text{ of } 360 = (5\% \times 9) \text{ of } 360$$

$$= \frac{9}{20} \times 360 = 162$$

$$(45\%) \times 2 \text{ of } \left(\frac{360}{2}\right)$$

$$= \frac{90}{100} \times 180 = 162$$

$$48\% \text{ of } 2500 = (4\% \times 12) \text{ of } 2500$$

$$= \frac{12}{25} \times 2500 = 1200$$

$$\left(\frac{48\%}{4}\right) \text{ of } (2500) \times 4$$

$$= \frac{12}{100} \times 10000 = 1200$$

$$48\% \text{ of } 350 = (4\% \times 12) \text{ of } 350$$

$$= \frac{12}{25} \times 350 = 168$$

$$\left(\frac{48\%}{4}\right) \text{ of } (350) \times 4$$

$$= \frac{12}{100} \times 1400 = 168$$

$$56\% \text{ of } 1250 = (4\% \times 14) \text{ of } 1250$$

$$= \frac{14}{25} \times 1250 = 700$$

$$\left(\frac{56\%}{4}\right) \text{ of } (1250) \times 4$$

$$= \frac{14}{100} \times 5000 = 700$$

$$24\% \text{ of } 625 = (4\% \times 6) \text{ of } 625$$

$$= \frac{6}{25} \times 625 = 150$$

$$\left(\frac{24\%}{4}\right) \text{ of } (625) \times 4$$

$$= \frac{6}{100} \times 2500 = 150$$

$$65\% \text{ of } 320 = (5\% \times 13) \text{ of } 320$$

$$= \frac{13}{20} \times 320 = 208$$

$$(65\%) \times 2 \text{ of } \left(\frac{320}{2}\right)$$

$$= \frac{130}{100} \times 160 = 208$$

$$75\% \text{ of } 160 = (25\% \times 3) \text{ of } 160$$

$$= \frac{3}{4} \times 160 = 120$$

$$(75\%) \times 2 \text{ of } \frac{160}{4}$$

$$= \frac{300}{100} \times 40 = 120$$

$$76\% \text{ of } 1250 = (4\% \times 19) \text{ of } 1250$$

$$= \frac{19}{25} \times 1250 = 950$$

$$\left(\frac{76\%}{4}\right) \text{ of } (1250) \times 4$$

$$= \frac{19}{100} \times 5000 = 950$$

Concept of BODMAS :-

B O D M A S
 (Bracket) (of) (Division) (Multiplication) (Addition) (Subtraction)

Basic (DMAS)

Q1. $84 \div 7 \times 5 \div 3 = ?$

(A) 20 (B) 18 (C) 14 (D) 30 (E) 10

$$\frac{84}{7} \times \frac{5}{3} = 20$$

Q2. $30 + 12 \times 2 = ?$

(A) 60 (B) 42 (C) 36 (D) 54 (E) 84

$$30 + (12 \times 2)$$

$$30 + 24 = 54$$

Q3. $60 - 24 \div 3 = ?$

(A) 56 (B) 52 (C) 48 (D) 36 (E) 44

$$60 - \left\{ \frac{24}{3} \right\} = 60 - 8 = 52$$

Q4. $45 \div 5 + 7 \times 2 = ?$

(A) 21 (B) 25 (C) 23 (D) 31 (E) 16

$$\left\{ \frac{45}{5} \right\} + \{ 7 \times 2 \}$$

$$9 + 14 = 23$$

Q5. $100 - 18 \times 3 + 24 \div 6 = ?$

(A) 72 (B) 46 (C) 50 (D) 64 (E) 40

$$\{100\} - \{18 \times 3\} + \left\{\frac{24}{6}\right\}$$

$$100 - 54 + 4 = 50$$

Mixed $\times \div + -$ (No Brackets)

Q6. $36 \div 3 \times 2 + 15 - 8 = ?$

(A) 23 (B) 31 (C) 39 (D) 27 (E) 19

$$= \left\{\frac{36}{3} \times 2\right\} + 15 - 8$$

$$= (12 \times 2) + 15 - 8 = 24 + 7 = 31$$

Q7. $18 + 12 \div 3 \times 2 - 5 = ?$

(A) 23 (B) 21 (C) 20 (D) 22 (E) 15

$$= 18 + \left\{\frac{12}{3} \times 2\right\} - 5$$

$$= 18 + 8 - 5 = 21$$

Q8. $50 - 6 \times 4 + 32 \div 8 = ?$

(A) 29 (B) 30 (C) 32 (D) 26 (E) 31

$$= 50 - (6 \times 4) + \left(\frac{32}{8}\right)$$

$$= 50 - 24 + 4 = 30$$

Q9. $25 + 9 \times 3 - 48 \div 6 = ?$

(A) 46 (B) 44 (C) 45 (D) 9 (E) 43

$$= 25 + (9 \times 3) - \left(\frac{48}{6}\right)$$

$$= 25 + 27 - 8 = 44$$

Q10. $80 \div 5 + 7 \times 6 - 10 = ?$

(A) 48 (B) 50 (C) 49 (D) 47 (E) 128

$$= \left\{\frac{80}{5}\right\} + \{7 \times 6\} - 10$$

$$= 16 + 42 - 10 = 48$$

Brackets ()

Q11. $(25 + 15) - 10 = ?$

(A) 30 (B) 20 (C) 25 (D) 35 (E) 40

$$(25 + 15) - 10$$

$$= 40 - 10 = 30$$

while solving questions,
Brackets will be
prioritized from
Inwards to outwards.

Q12. $50 - (18 + 12) = ?$

(A) 15 (B) 10 (C) 50 (D) 20 (E) 30

$$= 50 - (18 + 12)$$

$$= 50 - 30$$

$$= 20$$

Q13. $6 \times (8 + 7) = ?$

(A) 84 (B) 90 (C) 78 (D) 96 (E) 102

$$= 6 \times (8 + 7)$$

$$= 6 \times 15$$

$$= 90$$

Q14. $(50 - 26) \div 4 = ?$

(A) 6 (B) 5 (C) 4 (D) 8 (E) 10

$$= (50 - 26) \div 4$$

$$= \frac{24}{4} = 6$$

Q15. $5 \times (30 - 12) \div 3 = ?$

(A) 24 (B) 36 (C) 30 (D) 40 (E) 20

$$= 5 \times (30 - 12) \div 3$$

$$= 5 \times \frac{18}{3}$$

$$= 5 \times 6$$

$$= 30$$

Multiple Brackets [] { }

Q16. $80 - [20 + (15 - 5)] = ?$

(A) 60 (B) 50 (C) 40 (D) 55 (E) 45

$$= 80 - [20 + (15 - 5)]$$

$$= 80 - [20 + 10]$$

$$= 80 - 30$$

$$= 50$$

Multiple Brackets .

Prioritization

Inwards to outwards
→

Q17. $\{60 \div [5 \times (3 + 1)]\} = ?$

(A) 5 (B) 12 (C) 6 (D) 3 (E) 4

$$= \{60 \div [5 \times (3+1)]\}$$

$$= \{60 \div [5 \times 4]\}$$

$$= \frac{60}{20} = 3$$

Q18. $90 - \{30 - [12 + (8 - 3)]\} = ?$

(A) 77 (B) 65 (C) 73 (D) 85 (E) 67

$$= 90 - \{30 - [12 + 5]\}$$

$$= 90 - \{30 - 17\}$$

$$= 90 - 13$$

$$= 77$$

Q19. $[48 \div (6 - 2)] \times \{15 - (9 - 4)\} = ?$

(A) 60 (B) 80 (C) 120 (D) 100 (E) 140

$$= [48 \div 4] \times \{15 - 5\}$$

$$= 12 \times 10$$

$$= 120$$

Q20. $\{[100 - (40 + 10)] \div 5\} + 6 = ?$

(A) 20 (B) 14 (C) 18 (D) 15 (E) 16

$$= \{[100 - 50] \div 5\} + 6$$

$$= \left(\frac{50}{5}\right) + 6$$

$$= 10 + 6$$

$$= 16$$

Powers (Orders)

Q21. $2^3 + 5 = ?$

(A) 15 (B) 13 (C) 16 (D) 10 (E) 8

$$\begin{aligned} &= 2^3 + 5 \\ &= (2 \times 2 \times 2) + 5 \\ &= 8 + 5 = 13 \end{aligned}$$

Q22. $3^2 \times 4 = ?$

(A) 36 (B) 24 (C) 48 (D) 32 (E) 27

$$\begin{aligned} &= 3^2 \times 4 \\ &= (3 \times 3) \times 4 \\ &= 9 \times 4 \\ &= 36 \end{aligned}$$

Q23. $100 - 5^2 = ?$

(A) 70 (B) 80 (C) 85 (D) 75 (E) 95

$$\begin{aligned} &= 100 - 5^2 \\ &= 100 - (5 \times 5) \\ &= 100 - 25 \\ &= 75 \end{aligned}$$

Q24. $6 + 2^3 \times 3 = ?$

(A) 33 (B) 30 (C) 27 (D) 24 (E) 36

$$\begin{aligned} &= 6 + (2^3) \times 3 \\ &= 6 + (2 \times 2 \times 2) \times 3 \\ &= 6 + (8 \times 3) \\ &= 6 + 24 \\ &= 30 \end{aligned}$$

Q25. $(3^2 + 4^2) - 5 = ?$

(A) 18 (B) 22 (C) 25 (D) 15 (E) 20

$$= (3^2 + 4^2) - 5$$

$$= (9 + 16) - 5$$

$$= 25 - 5$$

$$= 20$$

Negative Numbers

Q26. $-12 + 5 \times 3 = ?$

(A) -15 (B) 15 (C) -3 (D) 8 (E) 3

$$= -12 + (5 \times 3)$$

$$= -12 + 15$$

$$= +3$$

Q27. $(-8 + 14) \times 3 = ?$

(A) 16 (B) 24 (C) 6 (D) 18 (E) 12

$$= (-8 + 14) \times 3$$

$$= +6 \times 3$$

$$= 18$$

Q28. $30 - [6 \times (-2)] = ?$

(A) 54 (B) 42 (C) 30 (D) 18 (E) 12

$$= 30 - [6 \times (-2)]$$

$$= 30 - [-12]$$

$$= 30 + 12$$

$$= 42$$

Q29. $(-24) \div 6 + (-5) \times 2 = ?$

(A) -14 (B) -6.5 (C) -4 (D) 14 (E) -16

$$\begin{aligned}
 &= \left(\frac{-24}{6} \right) + (-5 \times 2) \\
 &= (-4) + (-10) \\
 &= -4 - 10 = -14
 \end{aligned}$$

Q30. $50 - \{(-10) + 4^2\} = ?$

(A) 56 (B) 54 (C) 44 (D) 66 (E) 24

$$\begin{aligned}
 &= 50 - \{(-10) + 16\} \\
 &= 50 - \{-10 + 16\} \\
 &= 50 - (+6) \\
 &= 50 - 6 = 44
 \end{aligned}$$

Use of 'Of'

Q31. $24 \text{ of } 3 \div 2 = ?$

(A) 32 (B) 16 (C) 6 (D) 8 (E) 36

$$= (24 \times \frac{3}{2}) = 36$$

Q32. $36 \div 6 \text{ of } 3 = ?$

(A) 18 (B) 6 (C) 2 (D) 9 (E) 3

$$\begin{aligned}
 &= \underline{36} \div (6 \text{ of } 3) \\
 &= \frac{36}{6 \times 3} = \frac{36}{18} = 2
 \end{aligned}$$

Q33. $48 \div 4 \text{ of } 2 + 5 = ?$

(A) 11 (B) 29 (C) 24 (D) 19 (E) 9

$$= 48 \div (4 \text{ of } 2) + 5$$

$$= \frac{48}{4 \times 2} + 5 = \frac{48}{8} + 5 = 6 + 5 = 11$$

Q34. $60 \div 5 \text{ of } 3 - 4 = ?$

(A) 32 (B) 0 (C) 8 (D) 4 (E) 28

$$= 60 \div (5 \text{ of } 3) - 4$$

$$= \frac{60}{5 \times 3} - 4 = \frac{60}{15} - 4 = 4 - 4 = 0$$

Q35. $72 \div 3 \text{ of } 4 \div 2 = ?$

(A) 3 (B) 6 (C) 12 (D) 48 (E) 24

$$= 72 \div (3 \text{ of } 4) \div 2$$

$$= \frac{72}{3 \times 4} \times \frac{1}{2} = \frac{72}{12} \times \frac{1}{2} = 6 \times \frac{1}{2} = 3$$

Q36. $90 \div 6 \text{ of } 5 + 10 \div 2 = ?$

(A) 80 (B) 20 (C) 8 (D) 15 (E) 5

$$= 90 \div (6 \text{ of } 5) + \left(\frac{10}{2}\right)$$

$$= \frac{90}{30} + \frac{10}{2}$$

$$= 3 + 5 = 8$$

Q37. $120 \div 8 \text{ of } 3 + 6 \text{ of } 2 = ?$

(A) 57 (B) 17 (C) 5 (D) 45 (E) 12

$$= 120 \div (8 \text{ of } 3) + (6 \text{ of } 2)$$

$$= \frac{120}{24} + 12$$

$$= 5 + 12 = 17$$

Q38. $84 \div 7 \text{ of } 2 \times 3 = ?$

(A) 72 (B) 18 (C) 36 (D) 24 (E) 12

$$= 84 \div (7 \text{ of } 2) \times 3$$

$$= 84 \div 14 \times 3$$

$$= \frac{84}{14} \times 3 = 6 \times 3 = 18$$

Q39. $150 \div 5 \text{ of } 3 \text{ of } 2 = ?$

(A) 180 (B) 30 (C) 5 (D) 90 (E) 15

$$= 150 \div (5 \text{ of } 3 \text{ of } 2)$$

$$= 150 \div (5 \times 3 \times 2)$$

$$= \frac{150}{30} = 5$$

Q40. $96 \div 4 \text{ of } (3 + 1) = ?$

(A) 6 (B) 24 (C) 96 (D) 12 (E) 48

$$= 96 \div 4 \text{ of } (3 + 1)$$

$$= 96 \div 4 \text{ of } 4$$

$$= \frac{96}{4 \times 4} = \frac{96}{16} = 6$$

Miscellaneous

Q41. $80 - [20 + (15 - 5)] = ?$

(A) 35 (B) 50 (C) 47 (D) 40 (E) 58

$$\begin{aligned}
 &= 80 - [20 + 10] \\
 &= 80 - 30 \\
 &= 50
 \end{aligned}$$

Q42. $\{60 \div [5 \times (3 + 1)]\} = ?$

(A) 3 (B) 4 (C) 0 (D) 5 (E) 2

$$\begin{aligned}
 &= \{60 \div [5 \times (3 + 1)]\} \\
 &= \{60 \div [5 \times 4]\} \\
 &= \frac{60}{20} = 3
 \end{aligned}$$

Q43. $64 \div 2 \text{ of } 4 + 3 \text{ of } 5 = ?$

(A) 23 (B) 38 (C) 143 (D) 31 (E) 40

$$\begin{aligned}
 &= 64 \div (2 \text{ of } 4) + (3 \text{ of } 5) \\
 &= 64 \div 8 + 15 \\
 &= \frac{64}{8} + 15 = 8 + 15 \\
 &= 23
 \end{aligned}$$

Q44. $90 - \{30 - [12 + (8 - 3)]\} = ?$

(A) 77 (B) 92 (C) 73 (D) 67 (E) 65

$$\begin{aligned}
 &= 90 - \{30 - [12 + (8 - 3)]\} \\
 &= 90 - \{30 - [12 + 5]\} \\
 &= 90 - \{30 - 17\} \\
 &= 90 - 13 \\
 &= 77
 \end{aligned}$$

Q45. $[48 \div (6 - 2)] \times \{15 - (9 - 4)\} = ?$

(A) 120 (B) 170 (C) 125 (D) 95 (E) 130

$$= [48 \div 4] \times \{15 - 5\}$$

$$= \frac{48}{4} \times 10 = 120$$

Q46. $81 \div 3 \text{ of } 3 + 2^3 = ?$

(A) 35 (B) 17 (C) 89 (D) 27 (E) 26

$$= 81 \div (3 \text{ of } 3) + (2 \times 2 \times 2)$$

$$= 81 \div 9 + 8$$

$$= \frac{81}{9} + 8 = 9 + 8$$

$$= 17$$

Q47. $\{[100 - (40 + 10)] \div 5\} + 6 = ?$

(A) 16 (B) 11 (C) 23 (D) 21 (E) 14

$$= \{[100 - 50] \div 5\} + 6$$

$$= \left\{ \frac{50}{5} \right\} + 6$$

$$= 10 + 6 = 16$$

Q48. $60 \div [3 \times (4 + 1)] = ?$

(A) 4 (B) 3 (C) 9 (D) 8 (E) 1

$$= 60 \div [3 \times 5]$$

$$= 60 \div 15$$

$$= 4$$

Q49. 200 of $80 \div 4 \times 5 = ?$

(A) 100 (B) 196 (C) 180 (D) 150 (E) 104 (F) 20000

$$= (200 \text{ of } 80) \div 4 \times 5$$

$$= \frac{200 \times 80}{4} \times 5$$

$$= 20,000$$

Q50. $80 - \{6 \times [5 + (2 \times 3)]\} = ?$

(A) 9 (B) 12 (C) 19 (D) 10 (E) 14

$$= 80 - \{6 \times [5 + 6]\}$$

$$= 80 - \{6 \times 11\}$$

$$= 80 - 66$$

$$= 14$$

Q51. $[18 + 6 \times (7 - 4)] \div 3 = ?$

(A) 14 (B) 9 (C) 12 (D) 18 (E) 15

$$= [18 + 6 \times 3] \div 3$$

$$= (18 + 18) \div 3$$

$$= \frac{36}{3} = 12$$

Q52. $144 \div 6 \text{ of } 3 + 18 \div 3 \text{ of } 2 = ?$

(A) 11 (B) 20 (C) 84 (D) 75 (E) 14

$$= 144 \div (6 \text{ of } 3) + 18 \div (3 \text{ of } 2)$$

$$= \left(\frac{144}{18} \right) + \left(\frac{18}{6} \right)$$

$$= 8 + 3$$

$$= 11$$

Q53. $90 \div \{(12 - 7) \times 3\} = ?$

(A) 4 (B) 2 (C) 6 (D) 10 (E) 3

$$= 90 \div \{5 \times 3\}$$

$$= \frac{90}{15} = 6$$

Q54. $\{50 - [8 \times (4 + 2)]\} + 10 = ?$

(A) 19 (B) 14 (C) 13 (D) 18 (E) 12

$$= \{50 - [8 \times 6]\} + 10$$

$$= \{50 - 48\} + 10$$

$$= 02 + 10$$

$$= 12$$

Q55. $320 \div 8 \div 2 \div 5 = ?$

(A) 20 (B) 4 (C) 16 (D) 100 (E) None of these

$$= 320 \times \frac{1}{8} \times \frac{1}{2} \times \frac{1}{5}$$

$$= 4$$

Q56. $120 \times 3 \text{ of } 5 \div 30 \div 5 \text{ of } 2 = ?$

(A) 5 (B) 188 (C) 6 (D) -4 (E) 17

$$= 120 \times (3 \text{ of } 5) \div 30 \div (5 \text{ of } 2)$$

$$= 120 \times 15 \div 30 \div 10$$

$$= 120 \times 15 \times \frac{1}{30} \times \frac{1}{10}$$

$$= 6$$